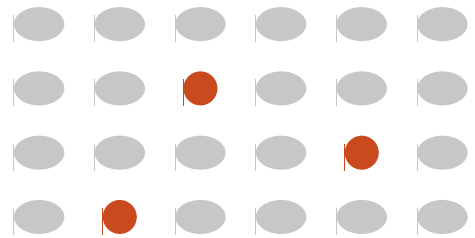


AZURE AI IN PRACTICE.



An evaluation of Azure AI services from the perspective of multi-cloud, open-source and local development.

**From platform
to architecture.**

PREPARED FOR

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A BRIEF ON AZURE AI IN PRACTICE

Introduction.

The Azure AI services covered in this module are extensive and well documented. The course introduces Machine Learning, Vision, NLP, Document Intelligence, Knowledge Mining, Generative AI and Azure AI Foundry as part of one broader Microsoft ecosystem. This document evaluates that ecosystem from outside it.

I completed the Whizlabs labs, but chose to do so in my own Azure subscription instead of relying on the shared course environment. That was a practical decision. It removed the dependency on limited three-hour Calendly slots, gave me control over the pace, and allowed me to experience Azure as an actual account owner rather than as a temporary lab user.

This document is not a summary of the course material. It is an evaluation of Azure AI as a platform from the perspective of someone who has built comparable solutions with AWS, Google Cloud, RunPod, open-source models and local hardware. From that perspective, Azure is one option among several. It is not automatically the right option.

The module was valuable because it clarified the Azure service hierarchy and the language Microsoft uses around AI capabilities. That matters. Many organisations already operate inside the Microsoft ecosystem, and being able to speak that language is useful in professional settings. However, understanding Azure also means understanding when it earns its place, when it creates unnecessary complexity, and when other architectures are more appropriate.

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SECTION 02

Azure AI in Context.

WHAT THE PLATFORM OFFERS AND HOW IT COMPARES

Azure AI is often presented as one integrated platform. In practice, it is a collection of services with different maturity levels, pricing models and strategic purposes. Some services are genuinely strong. Some are convenient wrappers around capabilities that also exist elsewhere. Some are valuable mainly because they integrate with the broader Microsoft enterprise stack.

That distinction matters. A business or technical team should not ask only whether Azure can solve a problem. In most cases, it can. The better question is whether Azure solves the problem better than the next-best alternative, and whether the additional cost, complexity and dependency are justified.

The right question is not "can Azure do this?" The right question is what Azure gives you that the next-best option does not.

Vision

Azure AI Vision provides pre-built capabilities for image analysis, including captions, tags, object detection and OCR. Azure Custom Vision adds the ability to train classification or detection models on user-provided data. Together, these services offer a low-barrier entry point into computer vision.

For generic use cases, this works well. The pre-built API can identify objects such as people, balls or vehicles. It can describe scenes and read printed text. For business users and teams without ML expertise, that is useful.

The limitation appears when the problem becomes domain-specific. Recognising basketball jersey numbers under match conditions, tracking individual players across video frames, handling occlusion, or mapping player positions onto a court are not solved by a generic object detection API. Custom Vision can help train a model, but it does not provide a full tracking pipeline. Identity persistence, OCR logic, geometric transformation and aggregation remain the responsibility of the developer.

In my own work, the alternatives are more direct. For training I use local hardware when possible, or RunPod when the workload requires more GPU capacity. The model is typically a fine-tuned YOLO variant for detection, combined with a tracker such as ByteTrack or BoT-SORT and a separate OCR step. Once trained, the model can run on local hardware, a Jetson device, Hetzner infrastructure or another cloud provider.

Document Intelligence

Azure AI Document Intelligence is one of the stronger services in the Azure AI catalogue. Its pre-built models for invoices, receipts and identity documents are accurate enough for many production use cases. The ability to train custom models from a small number of labelled examples is especially useful for structured forms with consistent layouts.

This is not a new problem space. In 2019 I worked on a receipt-scanning pipeline at Scoupy. Customers uploaded photos of shopping receipts, the backend extracted line items through OCR, and the system determined whether a cashback applied. That system existed before the current generation of managed AI services, but the same trade-offs were already visible.

Clean receipts are relatively easy. Edge cases are not. Faded thermal paper, folded receipts, unusual supermarket layouts, poor lighting and handwritten corrections all require additional rules, custom training or fallback logic. Azure Document Intelligence removes part of the engineering burden, but it does not remove the underlying complexity of the domain.

AWS Textract and Google Document AI offer comparable managed alternatives. Open-source options such as Donut, LayoutLMv3, Tesseract and layoutparser have also matured significantly. The strategic choice is therefore not simply about accuracy. It is a trade-off between engineering effort, operational control and ongoing per-page costs.

For a Dutch basketball club processing referee forms, the choice is not obvious. Azure Document Intelligence reduces development effort, but introduces recurring per-page costs. A local document model costs only compute time, but requires technical maintenance. Neither option is automatically right. The correct decision depends on volume, frequency, budget and who will operate the pipeline.

NLP

Azure AI Language Service provides sentiment analysis, named entity recognition, key phrase extraction and language detection. Azure Speech Service provides transcription and text-to-speech. Azure Translator supports translation across many languages.

These services are stable and useful, but they are also increasingly commoditised. Similar tasks can be performed with open-source models, Hugging Face transformers, spaCy pipelines, or direct API calls to frontier model providers such as OpenAI or Anthropic.

Azure's main advantage here is not always capability. It is integration. If an organisation already uses Azure infrastructure, Microsoft identity, Azure billing and Microsoft compliance tooling, then Azure Language Service can be a practical choice. It fits into an existing operational model. For standalone NLP problems, the choice is less obvious.

RAG and Azure AI Foundry

Azure AI Foundry positions itself as a platform for building retrieval-augmented generation applications. A typical Azure architecture combines Azure AI Search for retrieval, Azure OpenAI Service for generation and Foundry as the orchestration and evaluation surface.

This is a valid architecture. It offers a managed path for teams that want to build RAG applications quickly. It also provides strong alignment with Microsoft identity, monitoring, governance and procurement.

However, RAG is not fundamentally an Azure concept. A RAG system is a combination of document processing, indexing, retrieval, prompt construction, generation and evaluation. Those parts can be built inside Azure, but they can also be built outside it.

In a later module of this programme I built a RAG implementation outside Azure. It used a Python API, hybrid retrieval with BM25 and vector search through Chroma, page-aware chunking, contextual retrieval, an admin layer with metrics, and full containerisation. That implementation was assessed as production-grade.

The comparison is therefore concrete. The Azure route provides faster setup and managed scaling. The custom route provides more control over chunking strategy, retrieval weighting, evaluation and portability. For a team that needs an internal Q&A system quickly, Azure is reasonable. For a team that wants to own and optimise the retrieval layer, a custom architecture can be stronger.

An index built in Azure AI Search cannot simply be moved elsewhere without rebuilding it.

The portability question is important. A RAG application built tightly around Azure-specific services will gradually become operationally and economically dependent on those services. That dependency may be acceptable, but it should be a deliberate architectural decision rather than an accidental side effect.

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**AZURE SERVICE
AREAS EVALUATED**

3

**CLOUD PROVIDERS USED
IN PRODUCTION**

1

**ARCHITECTURE DECISION
THAT MATTERS MOST**

SECTION 03

From the Field.

USE CASE: PLAYER TRACKING FOR AMATEUR BASKETBALL

GPS-based player tracking is standard in professional football. Amateur indoor sports do not have the same access to this kind of analysis. GPS does not work reliably inside steel-framed sports halls. Basketball, futsal, handball and volleyball all share this limitation.

Computer vision offers an alternative. A camera positioned at one end of the hall can record the full court. In principle, the video can be used to identify where players move, how often they sprint, how much distance they cover, and how their positioning develops during a match.

This is a realistic technical problem, but not a trivial product problem. The challenge is not only whether the technology can work. The challenge is whether it can work under the constraints of amateur sport: low budgets, inconsistent video quality, volunteer-run operations and, most importantly, the fact that many players are minors.

The problem in scope

At BV Hoofddorp, a coach wants to understand where each player moved during a match. Useful outputs would include heatmaps per player, distance covered, time spent in offensive and defensive halves, and movement patterns over time. The footage already exists because many games are recorded for video review. What is missing is the analysis layer that turns video into player-level insight.

The technical pipeline is known:

- 1 Detect players in each frame.
- 2 Track players across frames, including moments where players overlap or cross paths.
- 3 Read jersey numbers to link tracks to player identity.
- 4 Map image coordinates onto court coordinates using homography.
- 5 Aggregate movement data into heatmaps and statistics.

None of these steps require new scientific research, but combining them into a reliable pipeline under amateur recording conditions is still a serious engineering challenge. The main issue is not only technical accuracy. The main issue is whether the architecture is legally, economically and operationally defensible.

The Azure route

The Azure-based version of this system would likely require multiple services. Azure AI Vision could handle generic object detection. Azure Custom Vision could be used to fine-tune detection for basketball-specific imagery. Azure Machine Learning would be needed for the custom parts of the pipeline: tracking, jersey OCR, homography and aggregation.

That immediately changes the nature of the solution. Azure AI Vision alone does not maintain player identity across frames. Custom Vision can help with detection, but it does not solve tracking. The system would still require custom code. Azure would mainly provide hosting, managed services and integration around that code.

The second issue is cost. Azure Vision inference is billed per image. A single basketball match at 30 frames per second over 40 minutes of play time produces around 72,000 frames. Processing every frame through a managed vision API is not realistic for an amateur club. Even processing one frame per second still results in 2,400 inference calls per match. With multiple youth teams playing weekly, those calls add up quickly.

For a professional sports organisation, that cost might be acceptable. For an amateur association, it is difficult to justify when the output is a coaching aid rather than a core revenue-generating system. The third issue is governance.

Why Azure was rejected

The players are minors. Video footage of them is personally identifiable data. If the system tracks movement patterns and links those patterns to jersey numbers or named players, the sensitivity increases further. Depending on implementation and purpose, this can move into biometric or behavioural profiling territory under GDPR.

That does not automatically make the project impossible, but it raises the threshold significantly. A volunteer-run sports club would need a clear legal basis, transparent communication with parents and players, data minimisation, retention rules, access control, processor agreements and likely a Data Protection Impact Assessment. These are not minor administrative steps. They are part of the architecture.

Infrastructure choice is a governance choice.

The use of a US-based cloud provider adds another governance layer. EU data residency can reduce certain risks, but it does not remove the fact that US providers may be subject to US legal obligations. For sensitive data about Dutch minors, this matters.

The recent reporting that Microsoft shared documents containing names of employees of the Dutch ACM and Autoriteit Persoonsgegevens with a committee of the United States House of Representatives illustrates this jurisdictional complexity. The documents related to the Digital Services Act and the broader US political discussion around online speech. The issue is not that the technical storage location was wrong. The issue is that data handled by a US-based provider can become subject to legal and political processes outside the European context in which that data originated.

For an amateur sports club, this is not an abstract legal debate. If adult civil servants involved in European regulatory work can be exposed through cross-jurisdictional pressure, then a club should be cautious before placing footage and movement data of children in the same legal environment.

What I built instead

The alternative implementation runs locally. A YOLO model fine-tuned on basketball footage handles player detection. A tracker maintains identity across frames. Jersey OCR is handled as a separate step and only runs on detections that meet a confidence threshold. Court homography uses reference points on the court to translate video coordinates into court coordinates. Aggregation then produces player-level heatmaps and movement statistics.

The system was tested on NBA broadcast footage, processed on RunPod. That data is public, high-resolution, well-lit and free of the GDPR concerns that apply to Dutch youth players, which made it appropriate to run on rented cloud GPU capacity rather than local hardware. The pipeline works end-to-end on this material.

The current limitation is compute economics rather than capability. Real-time tracking of ten players is feasible on RunPod for short test runs, but renting cloud GPU for every match in a season is not realistic for an amateur club. Local hardware at a club's price point is not yet powerful enough for game-speed processing. That is the gap between a working prototype and a deployable product.

The architecture is sound. The privacy position is stronger than a hyperscaler-based setup. The remaining challenge is hardware, not conceptual feasibility. Possible next steps include batched post-match processing, a dedicated edge device such as a Jetson Orin, or a Dutch or European processor with a legal posture that fits the sensitivity of the data.

Comparison

AREA	AZURE ROUTE	LOCAL ROUTE
Functional coverage	Partial; requires Azure ML and custom code	Full custom pipeline
Time to first prototype	Faster initial setup	Slower start, full control after
Cost model	Per-use inference costs	Compute time and hardware costs
GDPR posture (minors)	Complex; needs strong legal justification	More defensible by design
Vendor dependency	High	Low
Long-term portability	Reduced over time	High

The Azure route is faster to start. The local route is more realistic once the actual constraints of amateur sport are included: limited budget, sensitive data, local accountability and long-term operation without unpredictable per-call costs.

SECTION 04

The Choice.

WHEN AZURE AI EARNs ITS PLACE AND WHEN IT DOES NOT

When Azure AI is the right choice

Azure AI earns its place in several situations.

Enterprise integration

Organisations already running Microsoft 365, Entra ID, Azure infrastructure and Microsoft security tooling gain practical benefits from staying inside the same ecosystem. Identity, access control, audit logging, procurement and compliance processes are already aligned.

Regulated procurement

Microsoft's compliance documentation and certifications can carry real weight in enterprise environments. ISO, SOC, HIPAA and similar certifications help organisations pass procurement and risk-management gates.

Teams without deep ML expertise

Pre-built services can shorten development time for common use cases such as invoice extraction, sentiment analysis, translation, speech-to-text or basic internal Q&A. The managed platform hides a lot of complexity, which can be valuable.

Operational alignment

In these contexts, Azure is not just technically capable. It is organisationally convenient. That convenience has value, and dismissing it would be naive.

When Azure AI is the wrong choice

Azure AI is a poor fit when the organisation needs high control, low cost and long-term portability.

01 / SENSITIVE DATA

Especially involving children, health, surveillance or movement patterns

The issue is not only whether Azure has security controls. It does. The issue is whether the full legal, governance and jurisdictional picture is defensible for the organisation using it.

02 / SMALL ORGANISATIONS

Solo operators and limited budgets

Azure can become expensive and complex faster than expected. Costs that look acceptable in a demo can become painful at production volume. For many small teams, open-source models on rented or owned hardware provide better control and lower long-term cost.

03 / LONG-TERM PROJECTS

Architectures expected to last for many years

Managed services create dependencies. Azure AI Search indexes, Azure OpenAI deployments, Foundry workflows and service-specific configurations are not easily portable. Over time, the architecture can become shaped around Microsoft's abstractions, pricing and availability.

The deeper question

Most organisations do not choose Azure after a neutral comparison with alternatives. They choose Azure because Microsoft is already present. The Office licences are there. Active Directory is there. Procurement already knows the vendor. The support contract exists. The path of least resistance points toward Azure.

That is not automatically wrong. Existing operational fit is a legitimate factor. But it should not be mistaken for a full strategic evaluation.

The course material understandably assumes Azure. It is a module about Azure. The labs assume Azure. The architectures assume Azure. The service names and workflows assume Azure. That is appropriate for learning the platform. Professional decision-making starts after that. It starts when Azure is compared to the next-best option.

For my own work, that comparison is explicit. ClubDuty runs on Hetzner with PostgreSQL and a Node backend, because the operational profile of a small SaaS does not justify hyperscaler pricing. Other AI workflows use direct API access to OpenAI or Anthropic because the value lies in flexible model choice rather than managed cloud wrapping. Local fine-tuning runs on owned hardware or rented GPU capacity because the workload is bursty and data control matters.

Azure does not appear in those stacks. Not because Azure cannot work, but because it does not currently earn its place against the alternatives.

THE BETTER QUESTION

What does Azure give us that the next-best option does not, and is that difference worth the cost, complexity and dependency it creates?

That question produces better architecture decisions. It also produces honest ones, because it forces the organisation to articulate what it actually values.

01

DECISION-SUPPORT,
NOT DEFAULT

02

COMPARISON BEFORE
COMMITMENT

03

GOVERNANCE AS PART
OF ARCHITECTURE

SECTION 05

Closing Observation.

TECHNOLOGY CHOICES ARE GOVERNANCE CHOICES

Responsible AI is often discussed at the model level: fairness, transparency, human oversight, explainability and harm reduction. These are important. But responsibility also exists at the infrastructure level.

Where data lives matters. Which legal regime applies matters. Which company processes the data matters. Which dependencies are created matters. These choices shape what is possible, what is defensible and what risks are transferred to users.

The Toeslagenaffaire showed what can happen when algorithmic systems are treated as neutral facts instead of fallible tools. The ACM and AP disclosure illustrates how European governance data can become exposed to non-European legal and political processes when infrastructure and jurisdiction are not aligned. These are different cases, but they point toward the same broader lesson.

Technology choices are governance choices.

This document is not an argument against Azure. It is an argument against treating Azure as a default. The course delivered useful Azure knowledge. The professional task is to place that knowledge next to the alternatives and make a deliberate choice.

Azure is powerful. Azure is often useful. Azure is sometimes the right answer.

But it should still have to earn its place.

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About the Author.



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This document was written by Vincent Blokker as part of Module 3 of the AI Professional programme at the Amsterdam Data Academy. It evaluates Azure AI as a platform from the perspective of multi-cloud, open-source and local development practice.

The work reflects an applied perspective on responsible architecture, combining technical evaluation, economic context and governance design. Infrastructure choice is treated as a deliberate decision, not a default.

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AI Professional	B08 / Module 3	2026